COLLEGE CANTEEN MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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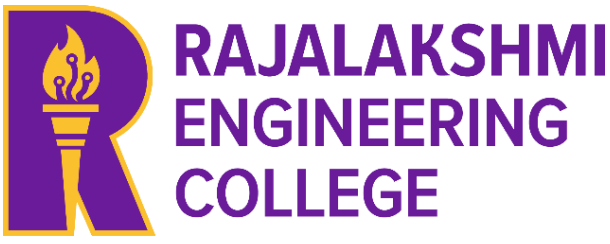
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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "COLLEGE CANTEEN MANAGEMENT SYSTEM" is the bonafide work of SANKARA GOMATHI R (2116240701470), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SANKARA GOMATHI R (2116240701470), hereby declare that the project report titled "COLLEGE CANTEEN MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, Deluge scripts, workflows and Blueprint — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11 November 2026

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SANKARA GOMATHI R

ABSTRACT

College canteens serve a large number of students and staff every day, yet many still depend on handwritten order slips, cash registers and separate stock notebooks. These methods make it hard to see which food items are available, what stage an order has reached, how much has been paid and when stock is running low. To address these challenges, the project "College Canteen Management System" is developed using Zoho Creator, a low-code application development platform.

The application consists of eight forms: Customer Profile, Food Category, Food Type, Food Menu, Food Order, Order Item, Payment Record and Inventory Transaction. The forms are linked through lookup fields, so that a menu item belongs to a category and a type, an order belongs to a customer, order items belong to an order and a menu item, and payments belong to an order. Fourteen reports (eight list reports and six Kanban reports) organise the stored data, and a dashboard presents summary panels, availability counts, quick links, two charts and an embedded Food Menu report. Eleven form workflows written in Deluge script generate customer codes and item numbers, fetch food prices, calculate line totals and order bills, reject failed payments and send e-mail notifications for new orders, payments and low stock. One scheduled workflow for low-stock alerts is configured but kept inactive. A Blueprint named Food Order Lifecycle defines the stages of an order, and permission profiles such as Canteen Manager and Inventory Controller restrict access to application data.

The application was evaluated using sample customers, menu items, orders, order items and payment data. The configured forms, reports, dashboard, workflows and Blueprint show how a low-code platform can bring menu management, ordering, billing, payments and inventory into a single application.

Keywords: *Low-Code Development; Zoho Creator; College Canteen; Order Management; Deluge Scripting; Workflow Automation; Inventory Tracking.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Canteen Management System
Application Domain	Food service and canteen management
Student Name & Register No.	SANKARA GOMATHI R – 2116240701470
Zoho Creator Edition	Trial edition (as shown in the application header)
Application Name	College Canteen Management System
Development / Live URL	https://creatorapp.zoho.in/sankaragomathi.r.2024.cse883/college-canteen-management-system1/#Section_1
Demo Credentials	Admin: sankaragomathi.r.2024.cse@rajalakshmi.edu.in
Number of Forms	8
Number of Reports (by type)	8 List Reports and 6 Kanban Reports (Total: 14)
Number of Pages / Dashboards	1 dashboard (Dashboard)
Number of Workflows	11 form workflows, 1 scheduled workflow (inactive), 1 Blueprint
Deluge Scripts	12 scripts (11 form workflows and 1 schedule) / 0 custom functions
Integrations	None external; built-in Send Mail action
Roles & Profiles Defined	CEO role; Canteen Manager and Inventory Controller profiles, along with the default Administrator, Read, Write and Developer profiles
Environment Used	Zoho Creator Development environment
Date of Production Deployment	Not deployed to production

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CSE	Computer Science and Engineering
DS	Deluge Script (script identifier)
ER	Entity–Relationship
FR	Functional Requirement
HoD	Head of the Department
INR	Indian Rupee
NFR	Non-Functional Requirement
TC	Test Case
UAT	User Acceptance Testing
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

A college canteen is a daily service point for hundreds of students and staff. In many colleges, orders are taken verbally or on paper, prices are added up by hand, payments are collected in cash, and stock levels are checked by looking at shelves. As the number of customers grows, these manual practices lead to long queues, billing mistakes and food running out without warning.

This project addresses that domain through the College Canteen Management System, a low-code application developed on Zoho Creator. The application brings customer profiles, the food menu, orders, order items, payments and inventory transactions into one platform. It uses Deluge workflows, structured reports and a dashboard to automate calculations and to give canteen staff a clear view of daily operations.

1.2 Problem Statement

Canteen data is usually spread across order slips, billing notebooks, payment receipts and stock registers. This makes it difficult to know what is on the menu, what each customer ordered and whether it has been paid for.

- Customer, menu, order, payment and stock records are kept separately with no link between them.
- Item totals and order totals are calculated by hand, which leads to errors and delays.
- Order progress (pending, preparing, ready, completed or cancelled) is not tracked in one place.
- Staff are not warned promptly about new orders, payments or low stock.
- A single summary of sales, availability and order status is not readily available.

1.3 Objectives

- To design forms and a linked data model for customers, food categories, food types, menu items, orders, order items, payments and inventory transactions.
- To automate identifiers, price lookup, line totals and order totals using Deluge scripting in form workflows.

- To send automated e-mail notifications for new orders, payment confirmation and low stock.
- To define the order lifecycle as a Blueprint and to reject payment records with a failed status.
- To provide organised views through list and Kanban reports and an interactive dashboard.
- To configure permission profiles for the canteen manager and the inventory controller.
- To evaluate the forms, reports, dashboard and workflows in the Development environment.

1.4 Scope of the Project

In scope: eight forms; Deluge form workflows; one scheduled workflow; the Food Order Lifecycle Blueprint; list and Kanban reports; the Dashboard; permission profiles; and evaluation in the Development environment.

Out of scope: online payment gateway integration, automatic stock deduction when an order is placed, a customer-facing ordering portal, third-party API connectivity and pro-code extensions.

1.5 Justification for a Low-Code Approach

Low-code platforms allow complete applications to be built quickly with drag-and-drop builders and a small amount of scripting. The canteen application needs data-entry forms, reports, a dashboard, e-mail alerts and a status workflow, all of which are available in Zoho Creator. Deluge scripts handle the calculations that the builders cannot, such as summing line totals into an order total. Because Zoho hosts the platform, the effort is spent on the canteen workflow rather than on servers, databases and user-interface code.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In most college canteens, orders are written on slips or noted down by the counter staff, the bill is added up manually, and payment is taken in cash or through a separate UPI or card device. Stock is tracked in a notebook and checked by eye. Because each of these records is kept on its own, no one has a combined picture of menu availability, order progress, payments and stock. Figure 2.1 illustrates this process.

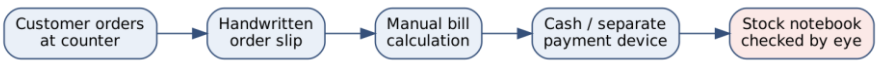


Figure 2.1: Existing Canteen Process

2.2 Limitations of the Existing System

- Records of customers, orders, payments and stock are stored separately.
- Bills and totals are calculated manually, which causes mistakes.
- Order status is communicated verbally and is easily lost.
- Low stock is noticed only after an item has run out.
- There is no dashboard or report for the manager to review.

2.3 Low-Code versus Traditional Development

Table 2.1 compares traditional development with the low-code approach for this application.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Development effort	Database schema, back-end logic, user interface and reports must be built from scratch.	Forms, reports and the dashboard are assembled in visual builders; Deluge handles calculations.
Skills required	Programming, database and web development.	Platform familiarity and basic Deluge.
Hosting	Developer manages servers and deployment.	Hosted by Zoho and accessed through a web link.
Workflow and status	State logic and notifications must be coded.	Blueprint and form-event workflows are configured visually.
Security and access	Authentication and authorisation must be coded.	Roles and permission profiles are provided by the platform.
Customisation limits	Unlimited flexibility.	Bounded by the platform; Deluge covers the required logic.
Cost	Developer time, server and licence costs.	Built on a Zoho Creator trial edition; a subscription is needed for continued use.
Maintenance	Code changes, testing and redeployment.	Changes are made in the builders or Deluge and tested in Development.

2.4 Literature Survey

The survey covers the official Zoho Creator and Deluge documentation and published references on low-code development. Table 2.2 summarises the reviewed literature.

Table 2.2: Summary of Literature Survey

S.No.	Author / Source	Title	Key Findings	Relevance to this Project
1	Zoho Corporation, online documentation	Zoho Creator Help Documentation [2]	Describes form fields, reports, workflows, Blueprint, schedules, roles and pages.	Primary reference for forms, reports, workflows and the dashboard.
2	Zoho Corporation, online documentation	Deluge Script Reference Guide [3]	Describes Deluge syntax, conditions, fetching records, loops and sendmail.	Basis for scripts DS-01 to DS-12.
3	Zoho Corporation, online learning material	Zoho Creator Learning Centre [4]	Learning material on platform features.	Supporting material for features used.
4	W. Rosam and O. Kalnichevski	Low-Code Development: A Practical Guide [1]	Introduces low-code concepts and platforms.	Supports the low-code justification.
5	D. Liu et al.	What's Wrong With Low-Code Development Platforms? An Empirical Study of Low-Code Development Platform Bugs	Identifies common bugs in low-code platforms, including design issues and unexpected behaviour.	Informs testing of workflows, calculations and report accuracy.

2.5 Emerging Trends in Low-Code Development

Low-code platforms make it practical to build domain-specific applications such as canteen management quickly. Visual builders, Deluge scripting, workflow automation and built-in analytics let a single developer deliver forms, reports, automation and dashboards in one environment.

2.6 Proposed System

The proposed College Canteen Management System keeps customers, menu items, orders, order items, payments and inventory in eight linked forms. Deluge scripts generate identifiers, fetch prices, calculate line totals and order totals, block failed payments and send e-mail alerts. A Blueprint defines the order lifecycle, and list reports, Kanban reports and a dashboard give the manager a live view of operations.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner (CEO role in the role hierarchy)	Manages all forms, reports, workflows and users.
Canteen Manager	Oversees daily operations	Manages customer profiles, menu, orders and payments; views inventory.
Inventory Controller	Tracks stock	Creates inventory transactions; views orders; updates menu availability.
Customer (student / staff)	Person who orders food	Represented by a Customer Profile record; receives e-mail confirmations.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Priority
FR-01	Register customers (students and staff) with an auto-generated customer code	High
FR-02	Maintain food categories, food types and the food menu with price and availability	High
FR-03	Record food orders linked to a customer	High
FR-04	Record order items linked to an order and a menu item, with price lookup and line total	High
FR-05	Calculate the order total from its order items	High
FR-06	Record payments against orders and reject failed payments	High
FR-07	Record inventory transactions and alert on low stock	Medium
FR-08	Send e-mail notifications for new orders and payments	Medium
FR-09	Show dashboard summaries, charts and reports	High
FR-10	Define the order lifecycle through a Blueprint	Medium
FR-11	Restrict access through permission profiles	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Readable forms, clear menu navigation and quick links on the dashboard
NFR-02	Integrity	Valid amounts, quantities and payment status; mandatory Order ID
NFR-03	Security	Permissions matched to role
NFR-04	Maintainability	Named forms, reports and workflows
NFR-05	Performance	Responsive reports at college-canteen data scale
NFR-06	Availability	Browser access subject to the platform subscription

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator trial / development builder
Client	Internet-connected laptop, desktop or tablet
Browser	Modern web browser
Scripting	Deluge editor
Data store	Zoho Creator built-in storage
External hardware	None required

3.5 Use Case Model

The canteen manager registers customers and maintains the menu. Orders are recorded with their items, payments are recorded against orders, and the manager reviews reports and the dashboard. The inventory controller records stock movements and monitors low-stock alerts.

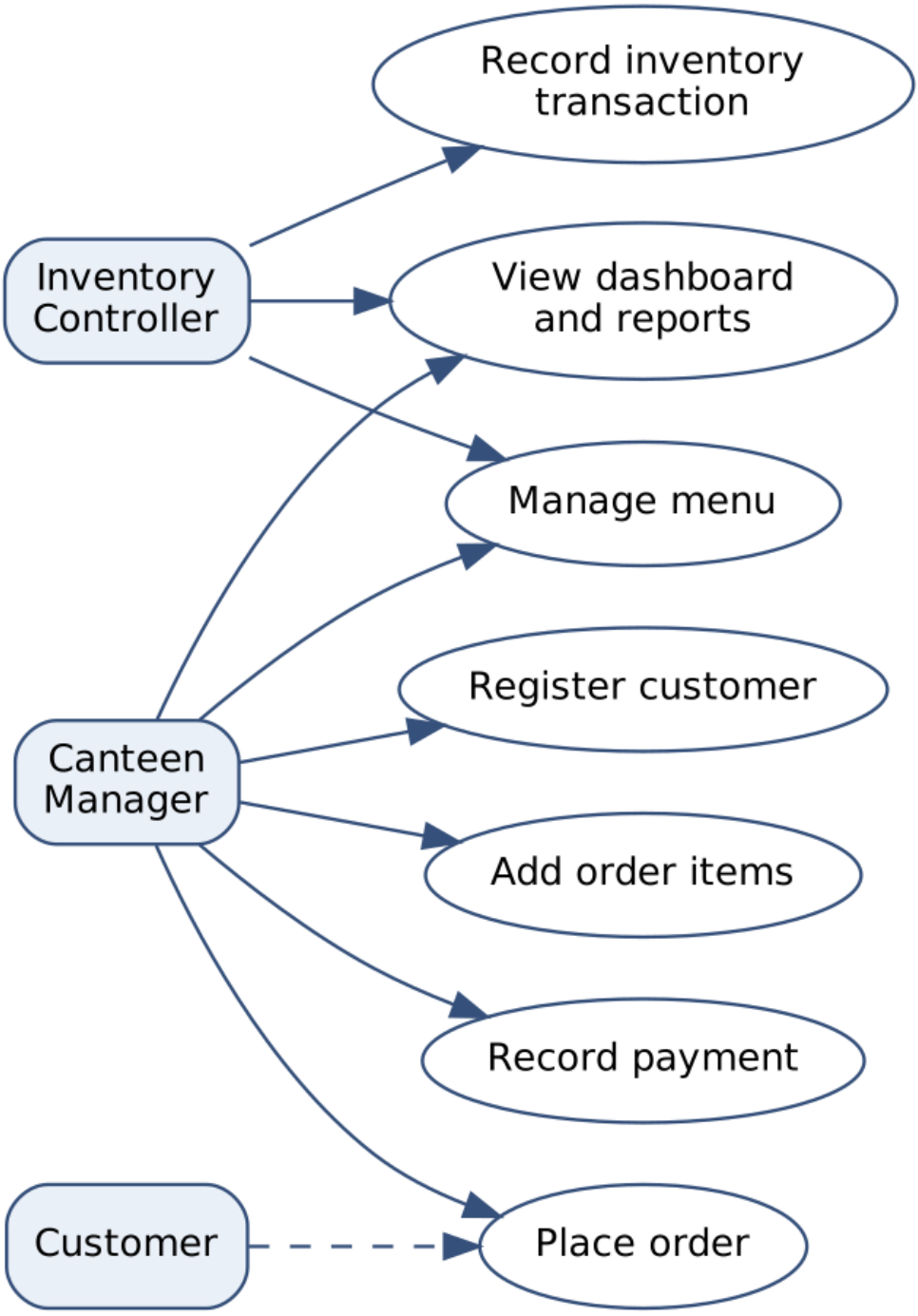


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Place and Pay for a Food Order

Attribute	Description
Use case	UC-01: Place and pay for a food order
Actors	Canteen Manager (on behalf of the customer)
Precondition	Customer profile and menu items exist
Main flow	Create order → add order items (price fetched, line total calculated) → order total updated → confirmation e-mail sent → record payment → update order status
Postcondition	Order, items and payment appear in their reports and on the dashboard
Exception	A payment with status Failed is rejected with an alert

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The browser interface connects to the Zoho Creator forms, reports, the dashboard page and the workflow configuration. Zoho Creator stores all records in its built-in data storage. Business logic is provided by Deluge form workflows, one scheduled workflow and a Blueprint. E-mail alerts use the platform's Send Mail action. No external database, REST API or webhook is used.

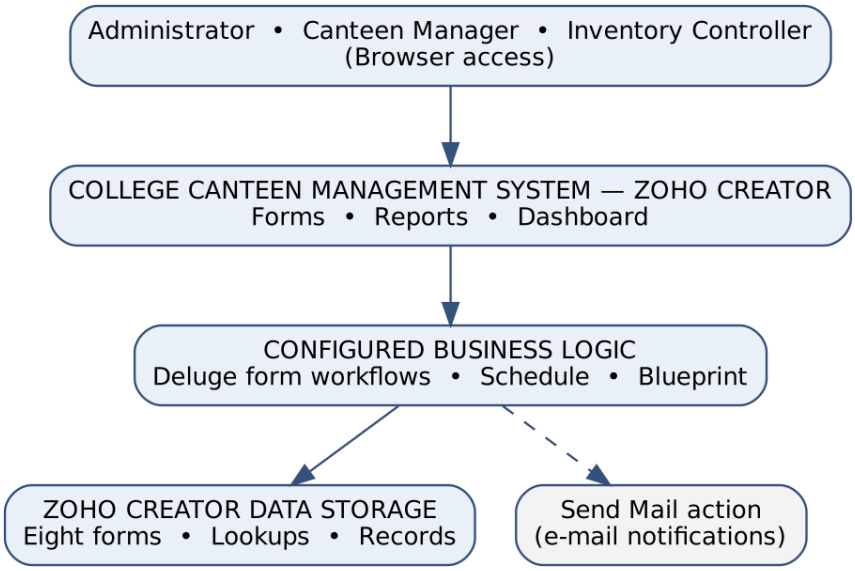


Figure 4.1: System Architecture of the Application

4.2 Data Model

The application has eight forms. Food Category and Food Type classify the Food Menu. Customer Profile identifies who orders. Food Order is the order header; Order Item holds the individual lines and links each order to a menu item. Payment Record links a payment to an order, and Inventory Transaction links a stock movement to a menu item.

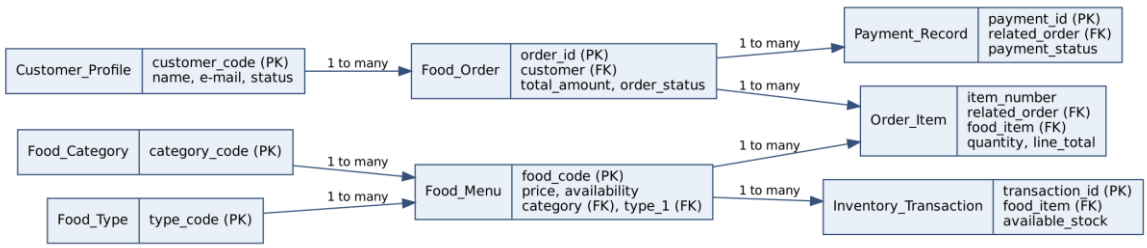


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity	Key information	Relationship
Customer Profile	Name, e-mail, phone, address, customer code, status, department, student/staff ID	One customer has many Food Orders
Food Category	Category name and code, vegetarian and spicy flags, allergen information	One category has many Food Menu items
Food Type	Type name and code, dietary category, preparation time	One type has many Food Menu items
Food Menu	Food name and code, price, availability, calories, featured flag	Looks up Food Category and Food Type; referenced by Order Item and Inventory Transaction
Food Order	Order ID, order date, total amount, status, payment method, delivery mode, estimated time	Looks up Customer Profile; has many Order Items and Payment Records
Order Item	Item number, unit price, quantity, line total, discount, tax amount	Looks up Food Order and Food Menu
Payment Record	Payment ID, amount, method, status, transaction reference, receipt URL	Looks up Food Order
Inventory Transaction	Transaction ID, type, quantity, available stock, minimum stock level, supplier, cost per unit	Looks up Food Menu

4.3 Form Design

The following tables list the fields of each form as configured in the application. Only the Order ID field of Food Order is marked mandatory in the application definition; other fields are optional.

Table 4.2: Field Specification – Customer Profile Form

Field Label	Link Name	Field Type	Remarks
Customer Name	customer_name	Name	First name shown; prefix, last name and suffix hidden
Email	email	Email	Contact e-mail used for order confirmation
Phone Number	phone_number	Phone Number	Contact number
Address	address	Address	Map coordinates captured
Registration Date	registration_date	Date	All days allowed
Customer Code	customer_code	Single Line	Auto-generated by workflow DS-01
Status	status	Drop Down	Active / Inactive / Pending
Department	department	Single Line	Student department
Student ID	student_id	Single Line	For students

Field Label	Link Name	Field Type	Remarks
Staff ID	staff_id	Single Line	For staff

Table 4.3: Field Specification – Food Category Form

Field Label	Link Name	Field Type	Remarks
Category Name	category_name	Single Line	Name of category
Description	description	Multi Line	Category details
Category Code	category_code	Single Line	Short code
Is Vegetarian	is_vegetarian	Checkbox	Default unchecked
Is Spicy	is_spicy	Checkbox	Default unchecked
Allergen Info	allergen_info	Multi Line	Allergen notes
Display Order	display_order	Number	Menu ordering
Status	status	Radio	Active / Inactive

Table 4.4: Field Specification – Food Type Form

Field Label	Link Name	Field Type	Remarks
Food Type Name	food_type_name	Single Line	Name of type
Description	description	Multi Line	Type details
Type Code	type_code	Single Line	Short code
Dietary Category	dietary_category	Drop Down	Gluten-Free / Veg / Non-Veg / Vegan
Preparation Time	preparation_time	Number	Preparation time
Is Available	is_available	Checkbox	Availability flag
Status	status	Radio	Active / Inactive

Table 4.5: Field Specification – Food Menu Form

Field Label	Link Name	Field Type	Remarks
Food Name	food_name	Single Line	Item name
Food Code	food_code	Single Line	Item code
Price	price	Currency (INR)	Selling price
Description	description	Multi Line	Item details
Availability Status	availability_status	Drop Down	Available / Out of Stock / Coming Soon
Category	category	Lookup	Food Category, with Add New
Type	type_1	Lookup	Food Type, with Add New
Image URL	image_url	URL	Picture link
Calories	calories	Number	Energy value
Is Featured	is_featured	Checkbox	Featured item flag

Field Label	Link Name	Field Type	Remarks
Status	status	Radio	Active / Inactive

Table 4.6: Field Specification – Food Order Form

Field Label	Link Name	Field Type	Remarks
Order ID	order_id	Single Line	Mandatory
Customer	customer	Lookup	Customer Profile, with Add New
Order Date	order_date	Date-Time	Order timestamp
Total Amount	total_amount	Currency (INR)	Updated by workflow DS-07
Order Status	order_status	Drop Down	Pending / Confirmed / Preparing / Ready / Completed / Cancelled
Payment Method	payment_method	Drop Down	Cash / Card / Online / Wallet
Special Instructions	special_instructions	Multi Line	Customer requests
Delivery Mode	delivery_mode	Radio	Pickup / Delivery
Estimated Time	estimated_time	Time	Expected ready time
Remarks	remarks	Multi Line	Staff notes

Table 4.7: Field Specification – Order Item Form

Field Label	Link Name	Field Type	Remarks
Item Number	item_number	Single Line	Set by workflow DS-03
Related Order	related_order	Lookup	Food Order
Food Item	food_item	Lookup	Food Menu
Unit Price	unit_price	Currency (INR)	Fetched from Food Menu (DS-05)
Quantity	quantity	Number	Ordered quantity
Line Total	line_total	Currency (INR)	Unit price × quantity
Item Status	item_status	Radio	Active / Removed
Discount	discount	Currency (INR)	Optional
Tax Amount	tax_amount	Currency (INR)	Optional
Notes	notes	Multi Line	Item notes

Table 4.8: Field Specification – Payment Record Form

Field Label	Link Name	Field Type	Remarks
Payment ID	payment_id	Single Line	Payment identifier
Related Order	related_order	Lookup	Food Order
Payment Date	payment_date	Date-Time	Payment timestamp
Amount	amount	Currency (INR)	Amount paid
Payment Method	payment_method	Drop Down	Cash / Card / Online / Wallet

Field Label	Link Name	Field Type	Remarks
Payment Status	payment_status	Drop Down	Pending / Success / Failed / Refunded
Transaction Reference	transaction_reference	Single Line	Gateway or receipt reference
Payer Name	payer_name	Name	Person paying
Remarks	remarks	Multi Line	Notes
Receipt URL	receipt_url	URL	Receipt link

Table 4.9: Field Specification – Inventory Transaction Form

Field Label	Link Name	Field Type	Remarks
Transaction ID	transaction_id	Single Line	Transaction identifier
Food Item	food_item	Lookup	Food Menu
Transaction Type	transaction_type	Drop Down	Inbound / Outbound / Return / Adjustment
Quantity	quantity	Number	Units moved
Transaction Date	transaction_date	Date-Time	Movement timestamp
Available Stock	available_stock	Number	Stock after movement
Minimum Stock Level	minimum_stock_level	Number	Reorder threshold
Remarks	remarks	Multi Line	Notes
Supplier Name	supplier_name	Name	Supplier
Cost Per Unit	cost_per_unit	Currency (USD)	Currency set to USD in the form

4.4 Process and Workflow Design

The main operational sequence is customer registration, menu maintenance, order creation, order-item entry, payment recording and inventory update. When a customer profile is added, a customer code is generated. When an order item is added, its number is generated, the unit price is fetched from the menu, the line total is calculated and the order total is recomputed from all items of that order. New orders and payments trigger e-mails, and an inventory transaction at or below the minimum stock level triggers a low-stock e-mail. A payment with status Failed is rejected at validation.

The Food Order Lifecycle Blueprint defines fourteen order stages and sixteen transitions, for example Submit Order for Payment, Confirm Payment, Start Order Processing, Mark Order as Ready and Cancel Order. Figure 4.3 shows the main path and the branches.



Figure 4.3: Workflow / Process Flow for the Order Lifecycle Blueprint

4.5 User Interface and Navigation Design

The left-hand menu lists Dashboard, Food Menus, Food Orders, Items, Inventory Transactions, Customer Profiles, Food Types, Payment Records and Food Categories. Each section holds its form and reports. Kanban reports give status-based views of menu availability, order status, inventory transaction type, customer status, dietary category and payment method.

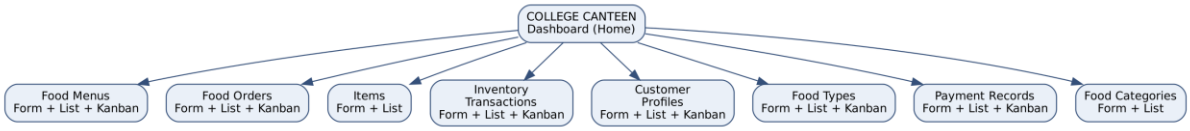


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

The application defines a CEO role (access to the data of all other users) and permission profiles. The Canteen Manager profile can create and manage customer profiles, food orders, food menu and payment records, and can view and modify inventory transactions. The Inventory Controller profile can create and manage inventory transactions, view and modify the food menu, and view food orders. Administrator, Read, Write and Developer are the default profiles.

Table 4.10: Role–Permission Matrix (as configured)

Component	Canteen Manager	Inventory Controller
Customer Profile	Create, view all, modify all, import, export	No access
Food Menu	Create, view all, modify all, import, export	View all, modify all
Food Order	Create, view all, modify all, import, export	View all
Payment Record	Create, view all, modify all, import, export	No access
Inventory Transaction	View all, modify all	Create, view all, modify all, import, export

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application "College Canteen Management System" was created from scratch in Zoho Creator and edited in the Development environment. The date format is dd-MMM-yyyy and the time format is 24-hour. The application runs on a Zoho Creator trial edition (the header showed 7 days remaining at the time of the screenshots). The left-hand menu is organised into sections for the dashboard and each form, as shown in Figure 5.1.

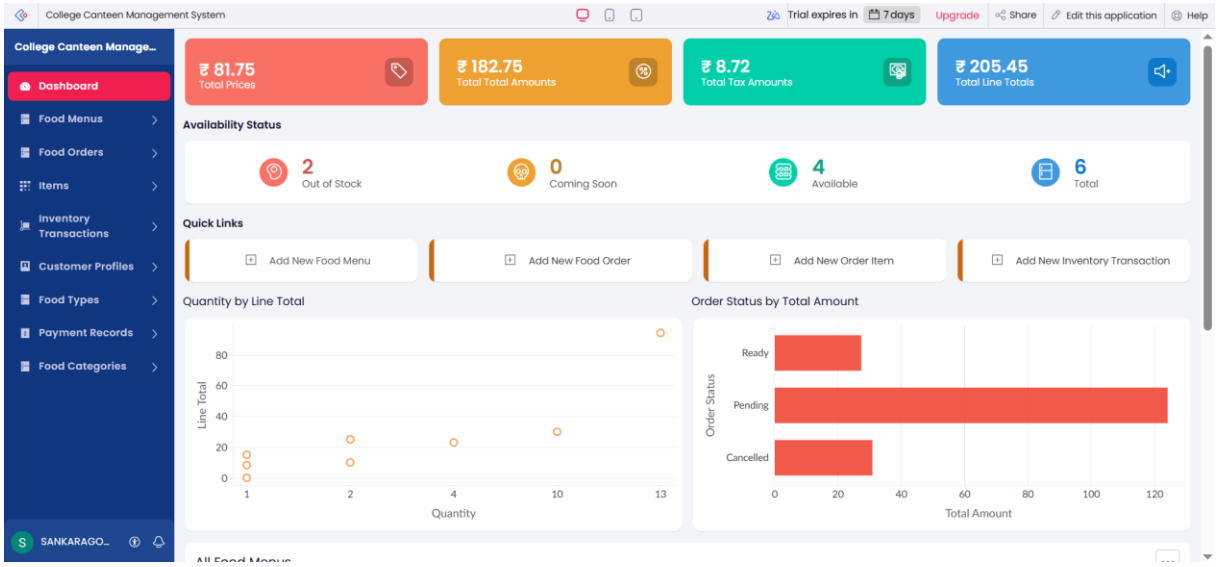
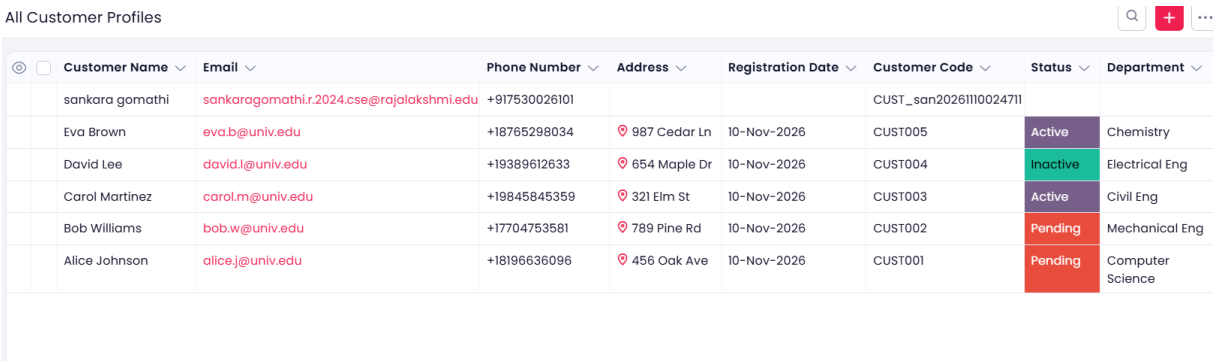


Figure 5.1: Dashboard and Navigation Sections in the Development Environment

5.2 Forms and Data Capture

Eight forms were built: Customer Profile, Food Category, Food Type, Food Menu, Food Order, Order Item, Payment Record and Inventory Transaction. The forms use single-line and multi-line text, name, e-mail, phone number, address, date, date-time, time, number, INR and USD currency, URL, drop down, radio buttons, checkbox and lookup fields. Lookup fields connect Food Menu to Food Category and Food Type, Food Order to Customer Profile, Order Item to Food Order and Food Menu, Payment Record to Food Order, and Inventory Transaction to Food Menu. Each form has Submit and Reset buttons on add, and Update and Cancel buttons on edit.

Sample data was entered to test the application: six customer profiles (five sample students plus the developer), six food menu items and seven food orders with their order items. Figure 5.2 shows the Customer Profile report with the auto-generated customer codes.



Customer Name	Email	Phone Number	Address	Registration Date	Customer Code	Status	Department
sankara gomathi	sankaragomathi.r.2024.cse@rajalakshmi.edu	+917530026101			CUST_san20261110024711		
Eva Brown	eva.b@univ.edu	+18765298034	987 Cedar Ln	10-Nov-2026	CUST005	Active	Chemistry
David Lee	david.l@univ.edu	+19389612633	654 Maple Dr	10-Nov-2026	CUST004	Inactive	Electrical Eng
Carol Martinez	carol.m@univ.edu	+19845845359	321 Elm St	10-Nov-2026	CUST003	Active	Civil Eng
Bob Williams	bob.w@univ.edu	+17704753581	789 Pine Rd	10-Nov-2026	CUST002	Pending	Mechanical Eng
Alice Johnson	alice.j@univ.edu	+18196636096	456 Oak Ave	10-Nov-2026	CUST001	Pending	Computer Science

Figure 5.2: All Customer Profiles Report

[Insert a screenshot of the Food Menu form (Add mode).]

Figure 5.3: Food Menu Form

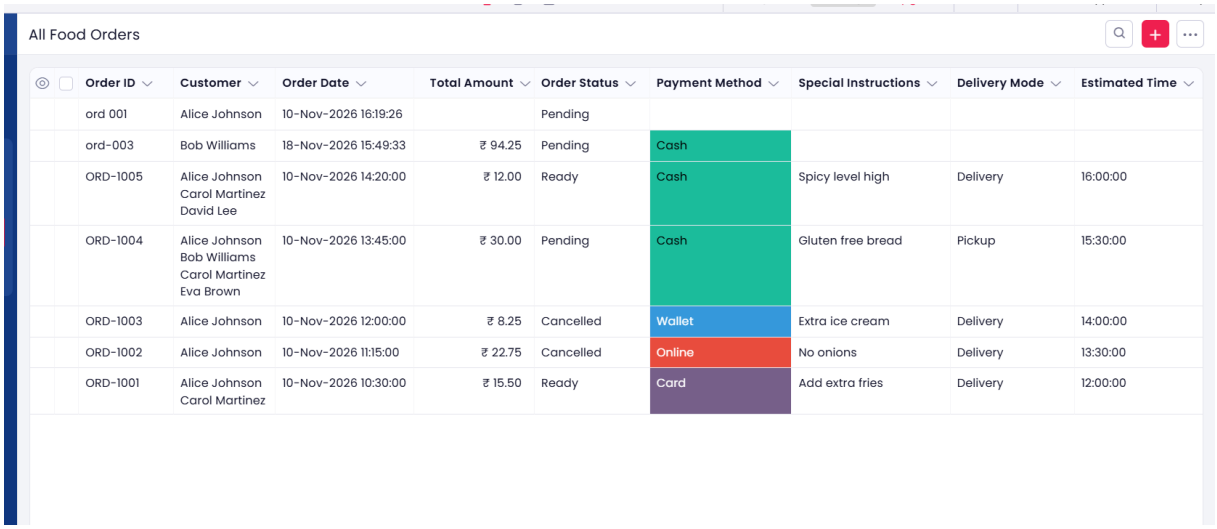
5.3 Reports

Eight list reports (one for each form) and six Kanban reports were created. The list reports apply conditional formatting to status fields, and the Food Menu list report shows the total, average, minimum and maximum of Price. Each report has quick view and detail view layouts; the Food Menu detail view also shows the related Food Type and Food Category records. Table 5.1 lists the reports.

Table 5.1: Reports Implemented

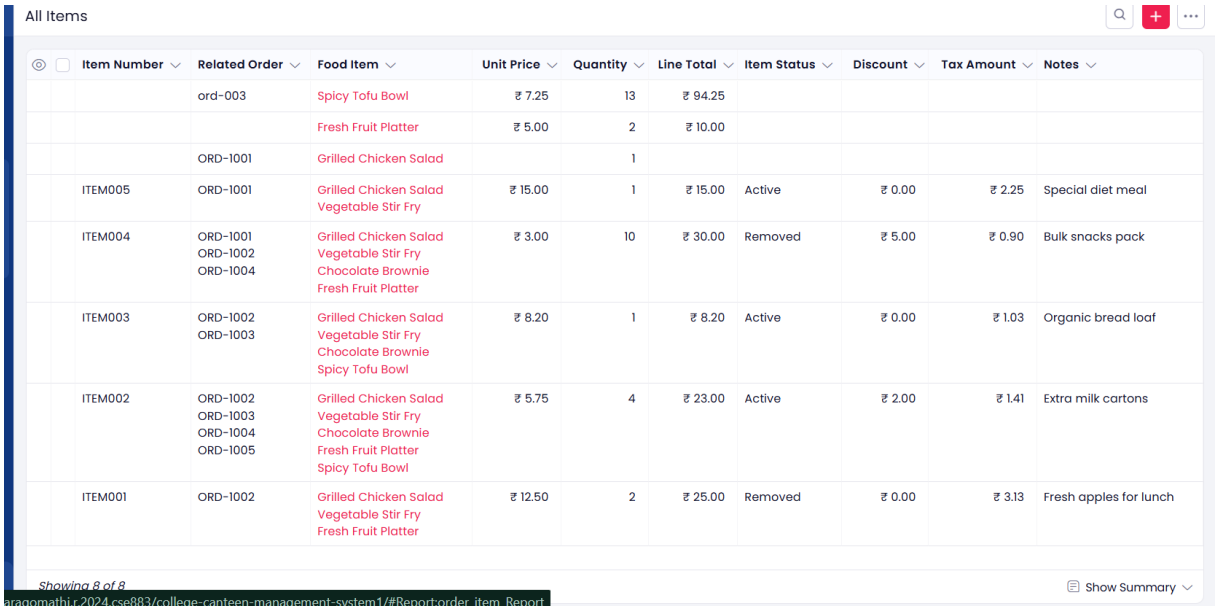
Report	Type	Source
All Food Menus	List	Food Menu
All Food Orders	List	Food Order
All Items	List	Order Item
All Inventory Transactions	List	Inventory Transaction
All Customer Profiles	List	Customer Profile
All Food Types	List	Food Type
All Payment Records	List	Payment Record
All Food Categories	List	Food Category
Availability Statuses	Kanban (grouped by Availability Status)	Food Menu
Order Statuses	Kanban (grouped by Order Status)	Food Order
Transaction Types	Kanban (grouped by Transaction Type)	Inventory Transaction
Statuses	Kanban (grouped by Status)	Customer Profile
Dietary Categories	Kanban (grouped by Dietary Category)	Food Type
Payment Methods	Kanban (grouped by Payment Method)	Payment Record

Figures 5.4 and 5.5 show the Food Order and Order Item reports.



Order ID	Customer	Order Date	Total Amount	Order Status	Payment Method	Special Instructions	Delivery Mode	Estimated Time
ord 001	Alice Johnson	10-Nov-2026 16:19:26		Pending				
ord-003	Bob Williams	18-Nov-2026 15:49:33	₹ 94.25	Pending	Cash			
ORD-1005	Alice Johnson Carol Martinez David Lee	10-Nov-2026 14:20:00	₹ 12.00	Ready	Cash	Spicy level high	Delivery	16:00:00
ORD-1004	Alice Johnson Bob Williams Carol Martinez Eva Brown	10-Nov-2026 13:45:00	₹ 30.00	Pending	Cash	Gluten free bread	Pickup	15:30:00
ORD-1003	Alice Johnson	10-Nov-2026 12:00:00	₹ 8.25	Cancelled	Wallet	Extra ice cream	Delivery	14:00:00
ORD-1002	Alice Johnson	10-Nov-2026 11:15:00	₹ 22.75	Cancelled	Online	No onions	Delivery	13:30:00
ORD-1001	Alice Johnson Carol Martinez	10-Nov-2026 10:30:00	₹ 15.50	Ready	Card	Add extra fries	Delivery	12:00:00

Figure 5.4: All Food Orders Report



Item Number	Related Order	Food Item	Unit Price	Quantity	Line Total	Item Status	Discount	Tax Amount	Notes
	ord-003	Spicy Tofu Bowl	₹ 7.25	13	₹ 94.25				
		Fresh Fruit Platter	₹ 5.00	2	₹ 10.00				
	ORD-1001	Grilled Chicken Salad		1					
ITEM005	ORD-1001	Grilled Chicken Salad Vegetable Stir Fry	₹ 15.00	1	₹ 15.00	Active	₹ 0.00	₹ 2.25	Special diet meal
ITEM004	ORD-1001 ORD-1002 ORD-1004	Grilled Chicken Salad Vegetable Stir Fry Chocolate Brownie Fresh Fruit Platter	₹ 3.00	10	₹ 30.00	Removed	₹ 5.00	₹ 0.90	Bulk snacks pack
ITEM003	ORD-1002 ORD-1003	Grilled Chicken Salad Vegetable Stir Fry Chocolate Brownie Spicy Tofu Bowl	₹ 8.20	1	₹ 8.20	Active	₹ 0.00	₹ 1.03	Organic bread loaf
ITEM002	ORD-1002 ORD-1003 ORD-1004 ORD-1005	Grilled Chicken Salad Vegetable Stir Fry Chocolate Brownie Fresh Fruit Platter Spicy Tofu Bowl	₹ 5.75	4	₹ 23.00	Active	₹ 2.00	₹ 1.41	Extra milk cartons
ITEM001	ORD-1002	Grilled Chicken Salad Vegetable Stir Fry Fresh Fruit Platter	₹ 12.50	2	₹ 25.00	Removed	₹ 0.00	₹ 3.13	Fresh apples for lunch

Figure 5.5: Order Item Report with Line Totals

5.4 Pages and Dashboards

The Dashboard page was built with the page builder. It has four coloured summary panels (Total Prices, Total Total Amounts, Total Tax Amounts and Total Line Totals), an Availability Status panel with counts for Out of Stock, Coming Soon, Available and Total, four quick-link buttons (Add New Food Menu, Add New Food Order, Add New Order Item and Add New Inventory Transaction), a scatter chart of Quantity against Line Total, a bar chart of Order Status against Total Amount, and an embedded All Food Menus report. At capture, the dashboard showed ₹81.75 as the sum of menu prices, ₹182.75 as the sum of order totals, ₹8.72 of tax, ₹205.45 of line totals, 2 items out of stock, 4 available and 6 in total (Figure 5.1).

5.5 Deluge Scripting

All application logic is written in Deluge inside form and schedule workflows. Twelve scripts were written; their complete source is in Appendix C. Table 5.2 summarises them.

Table 5.2: Deluge Scripts Implemented

ID	Script / Workflow	Form	Event	Purpose
DS-01	Auto-Generate Customer Code	Customer Profile	Add, on load	Builds code "CUST_" + first three characters of the user's e-mail + timestamp
DS-02	Validate Order Payment Status	Payment Record	Add or edit, on validate	Cancels submission with an alert if payment status is Failed
DS-03	Auto Generate Item Number	Order Item	Add, on success	Sets item number to "ITM-" + record ID
DS-04	Automatic Bill Calculation	Order Item	Add or edit, on success	Sets line total = unit price × quantity
DS-05	Fetch Food Price	Order Item	On user input of Food Item	Fetches the price from Food Menu into Unit Price
DS-06	Calculate Line Total	Order Item	Add, on user input of Quantity	Sets line total = unit price × quantity
DS-07	Calculate Final Order Bill	Order Item	Add or edit, on success	Sums line totals of all items of the order and updates the order's Total Amount
DS-08	Send Order Confirmation Email	Food Order	Add, on success	E-mails the customer that the order was placed
DS-09	Payment Confirmation Email	Payment Record	Add, on success	Confirms payment received (sent after the related order and customer are found)
DS-10	Notify Admin on New Order	Food Order	Add, on success	E-mails the administrator the order record ID and total
DS-11	Low Stock Alert	Inventory Transaction	Add or edit, on success	E-mails when available stock is at or below the minimum level
DS-12	Notify Low Stock Alert (schedule)	Inventory Transaction	Schedule on Transaction Date	Same alert as a scheduled workflow; currently inactive

[Insert a screenshot of one script (for example DS-07) in the Deluge editor.]

Figure 5.6: Deluge Script in the Workflow Builder

5.6 Workflows and Automation

Eleven form workflows and one schedule were configured (Table 5.3). A Blueprint named Food Order Lifecycle was configured on the Food Order form with fourteen stages and sixteen transitions, starting at Order Created (Section 4.4).

Table 5.3: Workflows Implemented

Workflow / process	Trigger or stage	Status
Auto-Generate Customer Code	Customer Profile created	Configured
Validate Order Payment Status	Payment Record created or edited	Configured
Auto Generate Item Number	Order Item created	Configured
Automatic Bill Calculation	Order Item created or edited	Configured
Fetch Food Price	Food Item selected in Order Item	Configured
Calculate Line Total	Quantity entered in Order Item	Configured
Calculate Final Order Bill	Order Item created or edited	Configured
Send Order Confirmation Email	Food Order created	Configured
Payment Confirmation Email	Payment Record created	Configured
Notify Admin on New Order	Food Order created	Configured
Low Stock Alert	Inventory Transaction created or edited	Configured
Notify Low Stock Alert	Schedule on Transaction Date	Configured, inactive
Food Order Lifecycle	Blueprint on Food Order	Stages and transitions configured

[Insert a screenshot of the Workflow page and of the Blueprint stages.]

Figure 5.7: Workflow List and Blueprint

5.7 Integrations

No external systems, REST APIs, webhooks or connections were used. The application uses two built-in platform capabilities, listed in Table 5.4.

Table 5.4: Integrations Implemented

Integration	Purpose	Status
Zoho Creator built-in storage	Persist form records	Used
Send Mail action	Order, payment and low-stock notifications	Configured in DS-08 to DS-12

5.8 User Management and Role-Based Access

A role named CEO was defined, whose users can access the data of all other users. Permission profiles were defined as follows: Canteen Manager (customer profiles, food

orders, food menu, payment records, inventory view and edit) and Inventory Controller (inventory transactions, food menu edit, food order view), in addition to the default Administrator, Read, Write and Developer profiles. Table 4.10 gives the permission matrix. Record-level access was not separately demonstrated with live sign-in of those users.

[Insert a screenshot of Settings > Users > Permissions / Roles.]

Figure 5.8: Roles and Permission Profiles

5.9 AI and Analytics Features

The dashboard and report views provide descriptive analytics (totals, counts and charts). No Zia predictive model, anomaly detector or automated AI recommendation was implemented.

5.10 Pro-Code Extension

No pro-code extension was developed. All logic is in Deluge inside workflows. Custom widgets, Catalyst functions or REST API integrations would become relevant only if the system had to exchange data with external services, such as an online payment gateway.

5.11 Mobile and Portal Access

Zoho Creator provides mobile and portal capabilities. The application header shows desktop, tablet and phone previews, but no tested mobile session or published portal login is evidenced for this project.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out in the Development environment using the sample data. Each check is marked by how it was confirmed: Observed (visible in a report or on the dashboard), Arithmetic verified (the displayed figures were recomputed by hand), or To be executed (the workflow must be run and its outcome recorded before final submission). No result in Table 6.1 is claimed unless it is marked Observed or Arithmetic verified.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected / Example	Evidence / Status
TC-01	FR-01	Customer profiles saved	Six customer records in the report	Observed
TC-02	FR-01	Customer code auto-generated	Code of the form CUST_ + prefix + timestamp (for example CUST_san20261110024711)	Observed
TC-03	FR-02	Menu availability counts	4 Available, 0 Coming Soon, 2 Out of Stock, 6 in total	Observed on dashboard
TC-04	FR-03	Food orders saved	Seven orders: 3 Pending, 2 Ready, 2 Cancelled	Observed
TC-05	FR-09	Pending order amount	₹94.25 + ₹30.00 = ₹124.25 (bar of about 124)	Arithmetic verified
TC-06	FR-09	Ready order amount	₹15.50 + ₹12.00 = ₹27.50	Arithmetic verified
TC-07	FR-09	Cancelled order amount	₹22.75 + ₹8.25 = ₹31.00	Arithmetic verified
TC-08	FR-09	Sum of order totals	94.25 + 30 + 12 + 8.25 + 22.75 + 15.5 = ₹182.75	Arithmetic verified
TC-09	FR-04	Unit price fetched and line total calculated	Line total = unit price × quantity	To be executed
TC-10	FR-05	Order total recomputed from items	Total Amount equals the sum of its line totals	To be executed
TC-11	FR-06	Failed payment rejected	Alert shown and record not saved	To be executed
TC-12	FR-08	Order and payment e-mails	E-mails delivered	To be executed
TC-13	FR-07	Low-stock alert	E-mail when available stock ≤ minimum level	To be executed

TC ID	FR Ref.	Test Description	Expected / Example	Evidence / Status
TC-14	FR-10	Blueprint transitions	Order moves from Order Created to Order Pending Payment and onward	To be executed
TC-15	FR-11	Profile restrictions	Inventory Controller cannot open Customer Profile	To be executed

6.3 Debugging Deluge Scripts

Each script is verified by submitting or editing a record and reading the calculated field in the report. Where a result is missing, the trigger event of the workflow (add, edit or user input) and the field link names used in the script are checked in the Workflow builder. Debug output or log screenshots were not captured, so no figure is included for this section.

6.4 Defect Log

The following issues were identified when the application definition and the dashboard were reviewed.

Table 6.2: Defect Log

Issue	Observation	Disposition
D-01	Payment Confirmation e-mail (DS-09) is addressed to a fixed administrator address instead of the customer's e-mail	Open; use customer_record.email as recipient
D-02	Low-stock schedule (DS-12) is inactive and uses "less than", while DS-11 uses "less than or equal"	Open; keep DS-11 active and align the condition
D-03	Cost Per Unit in Inventory Transaction is in USD while all other amounts are in INR	Open; change field currency to INR
D-04	Application time zone is America/Los_Angeles while the schedule uses Asia/Kolkata	Open; set the application time zone to Asia/Kolkata
D-05	Blueprint includes stages such as Shipped, Lost and Replacement Shipped that a canteen rarely needs	Open; trim stages to the canteen process
D-06	Stock is not reduced automatically when an order item is added	Open; listed as future enhancement
D-07	Dashboard panel "Total Prices" adds up menu prices and is not a sales figure	Open; replace with a count or an order-based figure
D-08	Sum of order totals (₹182.75) differs from the sum of line totals (₹205.45) because some sample orders carry manually entered totals and one has none	Open; recompute totals through the workflow

Issue	Observation	Disposition
D-09	DS-03 assigns Item_Number without the input. prefix	Open; confirm the field is updated and add the prefix

6.5 User Acceptance Testing

[Record here the feedback from the user who accessed the application, for example the invited reviewer's name, the permission profile used and the comments given. If UAT has not been carried out, state this clearly.]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Development environment of Zoho Creator on a trial edition. Stage and Production environments are available in the Environments section of the platform, but the application has not been published to Production in this project. The deployment checklist is given in Table 7.1.

Table 7.1: Deployment Checklist

Item	Evidence / status
Eight forms	Configured
Fourteen report views	Configured
Dashboard	Configured
Blueprint	Configured stages and transitions
Workflow execution	To be confirmed by running the test cases
Role access tests	Not yet performed
Production publish	Not done

7.2 Performance Optimisation

No formal performance measurements were taken because the data volume is small. One design choice is relevant to performance: DS-07 retrieves only the order items of the order concerned through a criteria-based fetch instead of reading the whole form.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Criteria-based fetch of one order's items	DS-07	Not measured	Not measured	Not measured

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy (CEO)	Application users	Unauthorised access to other users' data
Permission profiles	Canteen Manager, Inventory Controller	Users accessing modules outside their duty
Mandatory Order ID	Food Order form	Orders saved without an identifier

Security Control	Where Applied	Threat Mitigated
Failed-payment validation	Payment Record form (DS-02)	Orders recorded against a failed payment

Role and permission configuration is in place; enforcement for each profile was not independently validated with live sign-in.

7.4 Governance and Administration

Administration is handled by the application owner, who is the Super Admin of the Zoho Creator account. Workflows are given descriptive names so each rule can be found in the Workflow builder, and users are managed from the Zoho Creator administration console. The application definition was exported as a .ds file for backup.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The College Canteen Management System demonstrates centralised entry and display of canteen records. The sample outputs include six customer profiles, six menu items (four available and two out of stock), seven food orders with their order items, and a dashboard that summarises totals, availability and order status.

**Step 1: Customer registration

Customers are registered with their contact details and a code is generated automatically (Figure 8.1).

All Customer Profiles

Customer Name	Email	Phone Number	Address	Registration Date	Customer Code	Status	Department
sankara gomathi	sankaragomathi.r.2024.cse@rajalakshmi.edu	+917530026101			CUST_san20261110024711		
Eva Brown	eva.b@univ.edu	+18765298034	987 Cedar Ln	10-Nov-2026	CUST005	Active	Chemistry
David Lee	david.l@univ.edu	+19389612633	654 Maple Dr	10-Nov-2026	CUST004	Inactive	Electrical Eng
Carol Martinez	carol.m@univ.edu	+19845845359	321 Elm St	10-Nov-2026	CUST003	Active	Civil Eng
Bob Williams	bob.w@univ.edu	+17704753581	789 Pine Rd	10-Nov-2026	CUST002	Pending	Mechanical Eng
Alice Johnson	alice.j@univ.edu	+18196636096	456 Oak Ave	10-Nov-2026	CUST001	Pending	Computer Science

Figure 8.1: Scenario Step 1 – Customer Registration

**Step 2: Food order and order items

An order is recorded for a customer, and items are added to it. The unit price is taken from the menu, the line total is calculated and the order total is updated (Figures 8.2 and 8.3).

All Food Orders

Order ID	Customer	Order Date	Total Amount	Order Status	Payment Method	Special Instructions	Delivery Mode	Estimated Time
ord 001	Alice Johnson	10-Nov-2026 16:19:26		Pending				
ord-003	Bob Williams	18-Nov-2026 15:49:33	₹ 94.25	Pending	Cash			
ORD-1005	Alice Johnson Carol Martinez David Lee	10-Nov-2026 14:20:00	₹ 12.00	Ready	Cash	Spicy level high	Delivery	16:00:00
ORD-1004	Alice Johnson Bob Williams Carol Martinez Eva Brown	10-Nov-2026 13:45:00	₹ 30.00	Pending	Cash	Gluten free bread	Pickup	15:30:00
ORD-1003	Alice Johnson	10-Nov-2026 12:00:00	₹ 8.25	Cancelled	Wallet	Extra ice cream	Delivery	14:00:00
ORD-1002	Alice Johnson	10-Nov-2026 11:15:00	₹ 22.75	Cancelled	Online	No onions	Delivery	13:30:00
ORD-1001	Alice Johnson Carol Martinez	10-Nov-2026 10:30:00	₹ 15.50	Ready	Card	Add extra fries	Delivery	12:00:00

Figure 8.2: Scenario Step 2 – Food Order



Figure 8.3: Scenario Step 2 – Order Items and Line Totals

**Step 3: Monitoring on the dashboard

The dashboard shows totals, menu availability and the amount by order status (Figure 8.4).

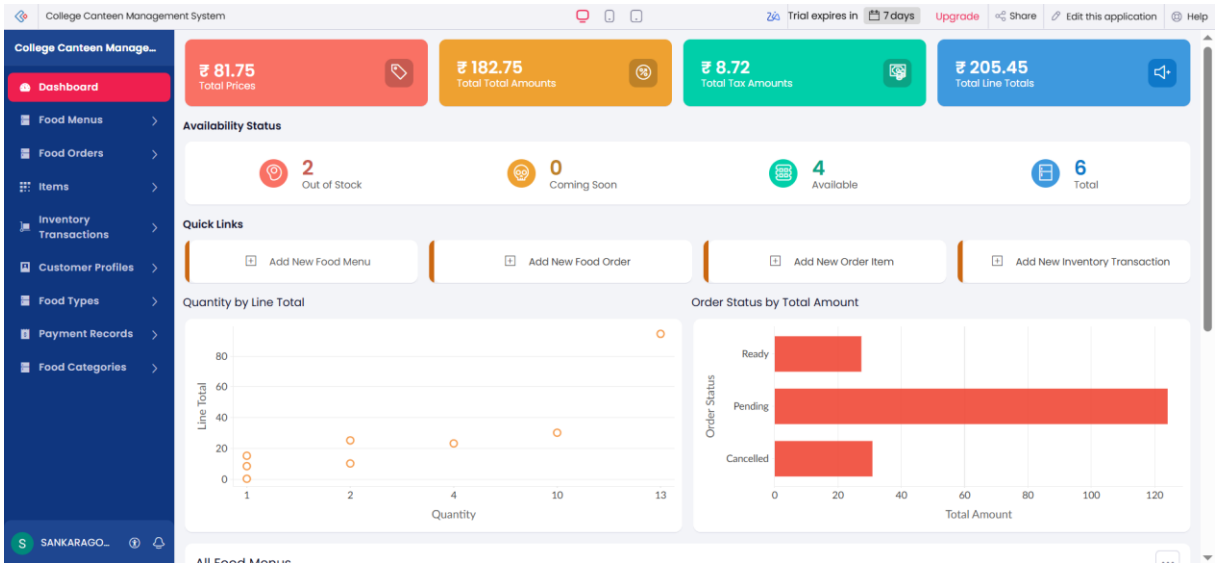


Figure 8.4: Scenario Step 3 – Dashboard Monitoring

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Observed outcome
Linked data model for eight forms	Eight forms with lookups configured
Automation of identifiers, price and totals	Scripts DS-01 to DS-07 configured; execution to be confirmed
E-mail notifications	DS-08 to DS-11 configured; delivery to be confirmed

Objective	Observed outcome
Order lifecycle and payment validation	Blueprint configured; failed-payment validation configured
Reports and dashboard	Fourteen reports and one dashboard configured and visible
Permission profiles	Canteen Manager and Inventory Controller profiles defined; access testing pending

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Criterion	Manual process	College Canteen Management System
Customer records	Not kept	Customer Profile with generated code
Menu and availability	Verbal / board	Food Menu with availability status
Billing	Manual addition	Line totals and order total calculated by workflow
Order tracking	Verbal	Order status, Kanban and Blueprint
Payments	Cash memo	Payment Record with status and validation
Stock	Notebook	Inventory Transaction with low-stock alert
Notifications	None	E-mail on new order, payment and low stock

No timing measurements were taken, so the improvements are stated qualitatively.

8.4 Limitations of the Low-Code Approach

The trial edition limits the time and number of users available for testing. Stock is not deducted automatically from orders, online payment is not integrated and the e-mail recipients are fixed addresses in several workflows. Some of these could be solved with additional Deluge code, and external payment integration would need a connection or REST API.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, Kanban views and the dashboard are no-code configuration. Deluge workflows for calculation, validation and e-mail, and the Blueprint, are low-code. A payment gateway integration, custom widgets or Catalyst functions would be pro-code extensions, which were not needed or demonstrated in this project.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Canteen Management System shows that a practical canteen application can be built on Zoho Creator with eight linked forms, fourteen reports, a dashboard, twelve Deluge scripts, an order Blueprint and permission profiles. The application brings together customers, menu, orders, items, payments and stock, and automates identifiers, price lookup, line totals, order totals and e-mail alerts. The workflow tests listed as "To be executed" should be run and recorded to confirm the automation fully.

9.2 Future Enhancements

The following are proposed enhancements and are not part of the completed work:

- Integrate an online payment gateway for card, online and wallet payments.
- Deduct stock automatically when an order item is added and restock on cancellation.
- Send payment and order confirmations to each customer's registered e-mail address.
- Offer a customer-facing ordering portal with a menu page and live order status.
- Simplify the Blueprint to the stages needed by a canteen and link it to the Order Status field.
- Extend role-based access and add further analytics, such as daily sales and best-selling items.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with forms and lookups, Deluge scripting, workflow automation, Blueprint design, reports and dashboards, permission profiles and testing of a low-code application. These skills are relevant to roles such as low-code developer and business-application developer.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/

[4] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/

[5] D. Liu et al., “What’s Wrong With Low-Code Development Platforms? An Empirical Study of Low-Code Development Platform Bugs.”

APPENDIX A
ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented; Partial = partly implemented, with the gap stated; N = not implemented in this project.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section
I	Application created and sections / menu organised	Y	Application with organised menu	5.1
II	Forms with 8 or more distinct field types	Y	Eight forms	5.2
II	Lookup field	Y	Menu, Order, Order Item, Payment, Inventory	4.2, 5.2
II	Subform	N	Order items kept in a separate form with lookup	4.2
II	Field validations / layout rules	Partial	Mandatory Order ID; payment status validation	5.2, 5.5
II	List report with filter, grouping, custom action	Partial	List reports with conditional formatting; filters and custom actions not configured	5.3
II	Kanban / Calendar / Timeline / Map report	Y	Six Kanban reports	5.3
II	Pivot table / chart report	N	Charts only on dashboard	5.4
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4
II	Deluge – form events	Y	On validate, on success, on user input, on load	5.5
II	Deluge – fetch, loops, collections	Y	Fetch and for each in DS-05, DS-07	5.5
II	E-mail / in-app notifications	Y	sendmail in DS-08 to DS-12	5.5
II	Form workflow	Y	Eleven form workflows	5.6
II	Scheduled workflow	Partial	One schedule, inactive	5.6
II	Approval process	N	Not used	5.6
II	Blueprint	Y	Food Order Lifecycle	5.6
II	Batch / report workflow	N	Not used	5.6
II	Testing and debugging evidence	Partial	Test cases listed; several to be executed	6
III	Integration with a Zoho service	N	Only built-in Send Mail	5.7
III	Connection / REST API call (invokeurl)	N	Not used	5.7

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section
III	Webhook or Creator REST API	N	Not used	5.7
III	Users, roles and permission profiles	Y	CEO role; Canteen Manager and Inventory Controller profiles	5.8
III	Record-level access criteria / portal	N	Not used	5.8
III	Zia / AI feature	N	Not used	5.9
III	Analytics (charts)	Y	Dashboard charts and summary panels	5.9
IV	Performance optimisation applied and measured	N	Not measured	7.2
IV	Deployment across environments	N	Development only	7.1
IV	Security controls	Partial	Permission profiles and validation	7.3
IV	Governance (audit trail, administration)	N	Not evidenced	7.4
V	Pro-code extension	N	Not used	5.10
V	Low-code limitations and spectrum analysis	Y	Discussed	8.4, 8.5
–	Mobile app / portal access	N	Not demonstrated	5.11

APPENDIX B
COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	4	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	4	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C
COMPLETE DELUGE SOURCE CODE

C.1 Auto-Generate Customer Code (DS-01)

```
if(input.ID.isNull())
{
    currentDateTime = now;
    currentUserEmail = zoho.loginuserid;
    emailPrefix = currentUserEmail.substring(0,3);
    timestamp = currentDateTime.toString("yyyyMMddHHmmss");
    customerCode = "CUST_".concat(emailPrefix).concat(timestamp);
    input.customer_code = customerCode;
}
```

C.2 Validate Order Payment Status (DS-02)

```
if(input.payment_status.equals("Failed"))
{
    alert(payment_status,"Cannot submit order with failed payment status. Plea
se resolve the payment issue first.");
    cancel submit;
}
```

C.3 Auto Generate Item Number (DS-03)

```
Item_Number = "ITM-" + input.ID.toString();
```

C.4 Automatic Bill Calculation (DS-04) and Calculate Line Total (DS-06)

```
input.line_total = input.unit_price * input.quantity;
```

C.5 Fetch Food Price (DS-05)

```
food_record = food_menu[ID == input.food_item];
input.unit_price = food_record.price;
```

C.6 Calculate Final Order Bill (DS-07)

```
total = 0;
for each item in order_item[related_order == input.related_order]
{
    total = total + ifnull(item.line_total,0);
}
order_record = food_order[ID == input.related_order];
if(order_record.count() > 0)
{
    order_record.total_amount = total;
}
```

C.7 Send Order Confirmation Email (DS-08)

```
sendmail
[
```

```
    from :zoho.adminuserid
    to :input.customer.email
    subject :"Your Canteen Order Confirmation"
    message :"Your order " + input.order_id + " has been successfully placed.
Thank you for ordering!"
]
```

C.8 Payment Confirmation Email (DS-09)

```
order_record = food_order[ID == input.related_order];
if(order_record.count() > 0)
{
    customer_record = customer_profile[ID == order_record.customer];
    if(customer_record.count() > 0)
    {
        sendmail
        [
            from :zoho.adminuserid
            to :"sankaragomathi.r.2024.cse@rajalakshmi.edu.in"
            subject :"payment confirmation"
            message :"your payment has been received successfully. thank you f
or using our college canteen!"
        ]
    }
}
```

C.9 Notify Admin on New Order (DS-10)

```
sendmail
[
    from :zoho.adminuserid
    to :"sankaragomathi.r.2024.cse@rajalakshmi.edu.in"
    subject :"new food order received"
    message :"a new food order has been placed. order record id: " + input.ID.
toString() + ". total amount: " + input.total_amount.toString()
]
```

C.10 Low Stock Alert (DS-11)

```
if(input.available_stock <= input.minimum_stock_level)
{
    sendmail
    [
        from :zoho.adminuserid
        to :"sankaragomathi.r.2024.cse@rajalakshmi.edu.in"
        subject :"low stock alert"
        message :"low stock alert! the available stock has reached the minimum
 stock level. please restock the item."
    ]
}
```

C.11 Notify Low Stock Alert – Schedule (DS-12, inactive)

```
if(input.available_stock < input.minimum_stock_level)
{
    emailSubject = "Inventory Alert: Low Stock";
    emailMessage = "Dear Team,\n\n";
    emailMessage = emailMessage.concat("The inventory has dropped below the mi
nimum level.\n");
    emailMessage = emailMessage.concat("Current Stock: ").concat(input.availab
le_stock.toString()).concat("\n");
```

```
    emailMessage = emailMessage.concat("Minimum Level: ").concat(input.minimum
_stock_level.toString()).concat("\n");
    emailMessage = emailMessage.concat("Transaction ID: ").concat(input.transa
ction_id);
    sendmail
    [
        from :zoho.loginuserid
        to :zoho.loginuserid
        subject :emailSubject
        message :emailMessage
    ]
}
```

APPENDIX D
USER MANUAL

D.1 Accessing the Application

Open the application link in a web browser (the Development URL is given in the Project at a Glance table) and sign in with the Zoho account that has access. The menu on the left lists the dashboard and the modules.

D.2 Instructions for the Administrator / Canteen Manager

- Open Customer Profiles and click the + button to register a customer. The customer code is generated automatically.
- Open Food Categories and Food Types to add classifications, then open Food Menus to add items with price, category, type and availability.
- Open Food Orders, enter a unique Order ID, choose the customer, date, payment method and delivery mode, and click Submit. A confirmation e-mail and an administrator e-mail are sent.
- Open Items (Order Item), choose the related order and the food item. The unit price is filled in automatically; enter the quantity. The line total and the order total are calculated.
- Open Payment Records to record the payment against the order. A payment with status Failed is rejected.
- Use the dashboard to view totals, menu availability and order status, and use the quick links to add new records.
- Use the Kanban reports to see records grouped by status and drag them between columns where editing is permitted.

D.3 Instructions for the Inventory Controller

- Sign in with an account that has the Inventory Controller profile.
- Open Inventory Transactions, choose the food item, the transaction type, the quantity, the available stock and the minimum stock level, and click Submit. A low-stock e-mail is sent when available stock is at or below the minimum.
- Open Food Menus to change an item’s availability status (Available, Out of Stock or Coming Soon).
- Open Food Orders to view incoming orders (view only).

APPENDIX E
EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SANKARA GOMATHI R	**Register No.**	2116240701470
Project Title	COLLEGE CANTEEN MANAGEMENT SYSTEM	**Date**	11.11.2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner
Name:_____

Signature of External Examiner
Name:_____

APPENDIX F
VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SANKARA GOMATHI R	**Register No.**	2116240701470
Department/section	CSE	**Date**	11.11.2026
Appliation URL verified	YES	**Demo Duration**	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent, G – Good, S – Satisfactory, NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name:_____ Name:_____